

Shailesh Dudala

Senior Applied AI / ML Engineer

Production Agentic AI · Predictive ML · Document Intelligence · MLOps · Healthcare & Insurance

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Applied AI and ML engineer with more than seven years across healthcare and insurance, from biomedical research and predictive modeling to production agentic systems. Turns ambiguous operational problems into complete intelligent workflows with measurable outcomes, reliable delivery, and explicit human authority.

PROFESSIONAL EXPERIENCE

Applied AI Engineer Consultant | MetLife via Bizintex

Jan 2026 - Present

- Builds production claims and document workflows spanning typed extraction, validation, exception routing, human review, evaluation, and observability.
- Supported approximately 90% lower handling effort and approximately 50% shorter time-to-claim-payable in a measured claims workflow.

Lead DS/ML Engineer Consultant | Inland Empire Health Plan via Infowave

Feb 2023 - Oct 2025

- Led predictive ML, calibration and explainability, local OCR/RAG, HEDIS evidence, FWA analytics, interoperability, and MLOps inside controlled data boundaries.
- Cleared a 7K-case backlog with approximately 90% lower review time; improved automated measure closures by 20% and reduced transportation waste by 18% in their respective workflows.

Lead Data Scientist / Product | Hexplora Predictive Analytics

Jun 2021 - Jan 2023

- Helped build and scale a 0-to-1 healthcare analytics platform across nine programs, combining calibrated risk models, data products, care-manager workflows, and reusable MLOps delivery.
- Supported \$500K in new revenue, approximately \$3M in broader client P4P impact, and approximately 50% faster model-deployment cycles.

Data Scientist | Hexplora Healthcare Analytics

Sep 2020 - May 2021

- Developed explainable readmission modeling, ETL automation, and the early product foundation for healthcare analytics delivery.

Data Scientist Consultant | CommonSpirit Health and Health New England via Infowave

Nov 2020 - May 2021

- Delivered hospital-operations and provider-data analytics spanning 142 hospitals and validation of 100K+ provider records; reduced manual provider review by 60% in a measured workflow.

Biomedical Informatics and Scientific Research | University of Chicago and AbbVie

Research foundation

- Worked across clinical sensors, genomics, scientific computing, biomedical research, and public-health modeling.

SELECTED CAPABILITIES

Agentic workflows and human review | Predictive ML, calibration, and explainability | OCR, RAG, and document evidence | FHIR/HL7 interoperability | Evaluation and observability | ML platforms and MLOps

SELECTED OPEN SOURCE

Meta Harness - evidence-gated execution and validation for coding-agent phases.

LLM Steering - local-first activation-steering experiments with reproducible artifacts.

EDUCATION + CREDENTIALS

M.S., Biomedical Informatics - University of Chicago, 2019

B.Tech., Computer Science and Engineering - SRM University, 2018

Yale Summer School, Public Health Modeling, 2019

AWS Certified Machine Learning Engineer - Associate | Azure AI Fundamentals | Power BI Data Analyst

RECOGNITION

Team recipient - Global HL7 AI Challenge, Transformative Impact in Healthcare Award (2025), Let's Talk Doc

PUBLICATION BOUNDARY: No phone, precise location, private application data, confidential payloads, PHI, PII, credentials, or production secrets.